

Smart Power Control System



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Here we describe an RF-based power control system using AVR microcontroller. This system provides three main facilities: automatic room power control, bidirectional visitor counter and automatic front-door light control. It will find application in office buildings and commercial buildings having separate entry and exit gates.

This automatic system will not only save electricity by automatically switching off the mains power line but also count the number of people entering and leaving the building. When everybody has left the building, the visitor count reaches zero and MCB output lines at the distribution board are automatically switched off. The automatic front-door light is another feature, which will be explained later.

Let's take the example of a multistory office building where there is only one guard at the main gate. There may be situations where:

1. The employees do not switch off electrical appliances like lights, fans and ACs before leaving the rooms.
2. The guard does not inspect the rooms and so does not switch off MCBs thinking that some workers may be still inside.

3. Burglars hidden inside steal office laptops and escape at midnight.

This automatic system provides a solution to all the above situations. Solution to the first situation is the automatic room-power control feature. The second and third situations can be prevented by alerting the guard through visitor counter system. With the counting system, the guard does not have to go and inspect each room, and he can remain posted outside. The counter indicates the exact number of people inside the building so the presence of any unauthorised person who crept in through the front gate and did not come out even after the office hours can be ascertained. The guard can go inside and check each room.

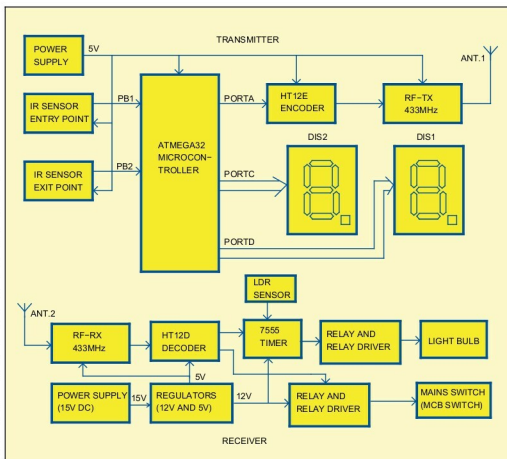


Fig. 1: Block diagram of smart power control system

Circuit and working

The block diagram of smart power control system is shown in Fig. 1. It has transmitter and receiver sections.

Transmitter. Fig. 2 shows the transmitter circuit. This section includes the AVR microcontroller, bidirectional counter, 7-segment displays and RF transmitter. The circuit operates off a 9V-12V DC supply. You can also use a suitable power adaptor. 7805 regulator is used to convert the supply to a regulated 5V power supply for the circuit. The glowing of LED1 indicates the presence of power in the circuit.

ATmega32 microcontroller. ATmega32 microcontroller is the heart of the transmitter section. It is a low-power